Harshit Singh

Software Engineer | React • Node.js/Express • REST APIs • SQL/NoSQL

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Education

Vellore Institute of Technology, Chennai — Chennai, India
B.Tech, Electronics and Computer Engineering — CGPA: ***7.97*** — *2023 – 2027*

New Era School (CBSE) — Ghaziabad, Uttar Pradesh
Class XII: ***87.16%*** *— Class X:* ***89.83%*** — *2020 – 2023*

Experience

AI & Robotics Intern — Autonomous Mobile Robotics [**Certificate**] [**GitHub**] — Jul 2025 – Aug 2025
STEMTEC AI & Robotics Technology Pvt. Ltd. — *Chennai, India*

- Engineered a **15 Hz** stochastic state-space estimation pipeline (Extended Kalman Filter, ROS 2 `robot_localization`) fusing wheel-encoder odometry with 6-DOF IMU telemetry, cutting mean spatial drift by **60%** (**0.45 m → 0.18 m**) and orientation error by **69.2%** (**7.8° → 2.4°**) across 10 m ground-truth trials.
- Built a real-time event-driven data pipeline processing **∼3,600 points/sec** via a publisher/subscriber architecture at **10 Hz**, applying 60° windowed stream filtering to drive a reactive control loop that achieved a **95% collision-free navigation success rate** (19/20 trials) at **sub-50 ms** end-to-end inference latency.

Projects

NeuroBridge AI — Full-Stack Remote Care Platform *(React, Node.js/Express, SQLite)* [**Live**] [**GitHub**]

- Designed and built a **70+ endpoint** RESTful API (Node.js/Express 5) backed by a **25-table** SQLite schema (WAL-mode concurrency) with JWT authentication and IP-level rate limiting, serving an **11-page** React 19 SPA.
- Engineered real-time WebRTC teleconsultation (Socket.IO, STUN/TURN) and a moderated community platform (**15 endpoints**, ML-assisted content filtering), plus automated PDF report generation and cron-scheduled notification pipelines.

TRIDENT — Unified Emergency Command Platform *(Flask, SQLite, Chart.js, Docker)*[**Demo**] [**Live**] [**GitHub**]

- Designed a RESTful API exposing **25 endpoints** over a **6-table** relational schema (**63 columns**, 4 foreign keys), powering a **2,960-line** single-page dashboard with **11** Chart.js visualizations and 30s real-time polling.
- Built a rule-based auto-dispatch engine scoring incidents across **15** weighted factors with atomic team assignment, and containerized the full stack (Docker, Gunicorn WSGI) for production deployment on Render.

GNOME Stocks Suite — Full-Stack Financial Data Platform *(Flask, SQLite, JavaScript, Render)* [**GitHub**]

- Built and deployed a **9-endpoint** REST API aggregating **3** external providers into a unified schema (normalizing **8** symbol formats), with a 3-tier caching architecture cutting repeat-request latency from **800ms+ to <1ms**.
- Developed an **832-line** vanilla JS SPA with interactive charting (**8** time ranges) and shipped automated CI/CD (GitHub Actions) with production deployment and keep-alive monitoring on Render.

Skills

Languages: JavaScript, Python, SQL, Java, C
Web & Backend: React, Node.js/Express, Flask, REST APIs, SQLite (Relational Schema Design)
Auth & Real-Time: JWT, bcrypt, Socket.IO, WebRTC
DevOps & Tools: Docker, CI/CD (GitHub Actions), AWS, Git, Linux (Ubuntu), Bash
ML/DL (Applied): PyTorch, TensorFlow/Keras, OpenCV, YOLOv8, MediaPipe
Professional Skills: Data-Driven Problem Solving, Cross-Functional Collaboration, Communication, Time Management

Achievements & Leadership

- **Top 5 Finalist, Smart India Hackathon (SIH) 2025** — SPARC, an edge ML multi-modal perception engine, selected from **500+ teams**. [**Demo**] [**Web-App**] [**Certificate**]
- **3rd Place, Technovation 2025** (Kurukshetra '25, CEG Guindy, Anna University) — ranked **2nd Runner Up out of 50+** teams; **INR 15,000** award. [**Certificate**]
- **Winner, Innovate X Impact (V-NEST)** — secured **1st Position out of 35** teams; **INR 5,000** award.
- **Deputy Operations Lead, Team Namo Nirvana (VIT Chennai)** — led **100+ member** teams across **5–6 competitions/semester**, boosting efficiency by **35%** and securing **2 sponsors.** [**LOA**] [**Certificate**] [**LOR**]

Languages

English (Fluent) | **Hindi** (Native)